A Calibrated In-Silico Screen for Corpus Necessity: Constructing the Parametric Ground Truth, and the Misled-Learner Blind Spot

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Abstract

Before running a human study on a corpus-bounded tutor, a designer wants to know something cheaper first: does the learner actually *rely* on the curated corpus, or is it coasting on what it already knows? We present an *in-silico measurement instrument* that answers this as a behavioral task outcome. The central quantity is *corpus necessity*: ablate a corpus item and measure the change in a task outcome scored against a reference policy—a with-versus-without contrast, not an influence score. Our headline move is to make necessity *trustworthy by constructing its ground truth*: we teach a fact into a model's weights and watch necessity fall as a *dose-response*. We calibrate the measure in a *simulator-free* fictional-fact question-answering task scored by an answer checker, where teaching invented facts drives necessity smoothly toward zero down the LoRA-scale gradient in two independent model families and three seeds each ($1.00 \rightarrow 0.48 \rightarrow 0.05$ for Qwen2.5-3B, $1.00 \rightarrow 0.93 \rightarrow 0.01$ for Phi-3.5-mini). A second, *decision*-based domain (a control-regret simulator with a hidden charted hazard) adds a caution that behavioral-necessity work should heed: a fact taught as prose becomes recallable yet does *not* move the scored decision—a *knowing* $\neq$ *doing* gap—unless the knowledge is *elicited* by reasoning, and it cannot be reliably *installed* by direct fine-tuning, which yields a shortcut policy rather than the conditional fact. On this calibrated measure we build a *corpus diagnosis* that labels each item *relied*, *redundant* (already known), *unusable* (present but the model fails), or *misled*. The sharp result is the last: a learner that faithfully follows a *wrong* retrieved rule registers as maximally reliant—the decision swings hard when the rule is ablated—yet is actively harmed, and standard necessity and faithfulness metrics, which only ask whether context influenced the output, are blind to it. Reading necessity as *decision-changes*$\times$*succeeds-with* separates this case out. We are explicit about scope: every result here is simulation-based mechanism evidence, not human-subject external validity. This is the pre-human-study screen; the human cohort study is the pre-registered endpoint it is built to de-risk, and is not a claim of this paper.

1 Introduction

A growing body of work builds *corpus-bounded* tutors and assistants: systems that abstain outside a curated corpus so that, in safety-critical procedural domains, a semantically plausible but instructionally wrong answer cannot be delivered. A designer who has built such a system faces a prior question, before any question about learning gains: *does the mechanism work?* Does the system actually rely on its corpus, and does the corpus actually carry the domain—or is the model answering from parametric memory and the corpus merely decorative? Answering this by human-subject study is slow, expensive, and confounded: a null there conflates a broken retrieval mechanism, an inadequate corpus, an unmotivated cohort, and a dozen other causes.

We argue that the *mechanism* question can be answered cheaply and in silico, and we build the instrument that does it. The core quantity is *corpus necessity*: remove a corpus item and measure how much a behavioral task outcome, scored against a reference policy, degrades. This is a with-versus-without *contrast*, not a measure of whether the context influenced the output. The difficulty is that a raw necessity reading is confounded—a near-zero value could mean the model already knew the fact, or that the without-context condition triggered some unrelated heuristic. Our contribution is to remove that confound *by construction*: we teach a fact into a model's weights along a graded schedule and show that necessity for that fact falls as a dose-response, giving the measure a known-groups calibration that influence-based methods structurally lack. On the calibrated measure we build a *corpus diagnosis* that labels each item relied / redundant / unusable / *misled*, and its sharpest output is the misled case: a learner faithfully following a wrong retrieved rule looks maximally reliant yet is harmed, invisible to standard metrics.

Our contribution is a measurement one, scoped as a pre-human-study screen. We make no learning-gains claim; every result is simulation-based mechanism evidence, and the human cohort study is pre-registered as the endpoint this screen de-risks, not something we report here. The paper proceeds as follows: Section 2 positions the measure against faithfulness and context-attribution; Section 3 defines the reference-policy instrument and its construct validity; Section 4 is the headline—necessity read as different-and-usable, calibrated by the construction dose-response; Section 5 turns the calibrated measure into the corpus diagnosis and the misled result; and Sections 6–7 scope and close. Appendices carry the categorical leakage verdict, a weight-level unlearning cross-check, estimator recovery, and a presence-vs-retrieval decomposition.

2 Related work

Faithfulness, context-attribution, and knowledge conflict. A large line asks whether a model uses provided context or parametric memory. Faithfulness metrics score whether an answer is grounded in the supplied context [Es et al., 2023]; context attribution ablates context sources and measures the change in output probability to attribute a generation to them [Cohen-Wang et al., 2024, Or, 2026]; knowledge-conflict work studies which side wins when context contradicts memory [Xu et al., 2024, Zhang et al., 2025]; leakage-free benchmarking filters items answerable with no context [Liu et al., 2026]; and intermediate-step scoring looks beyond the final answer [Jain and Vedam, 2026]. All of these measure whether context *influenced* an output, judged on answer text, with the corpus *present*. Our delta is on two axes. First, we measure *necessity*, a with-versus-without contrast—$\mathrm{acc}(\text{with corpus}) - \mathrm{acc}(\text{without corpus})$ —rather than influence: a faithfulness score is a function of the *(question, context, answer)* triple computed with the corpus present, and RAGAS never runs the without-corpus arm, so it is structurally blind to the quantity—a model that knows a fact and one that does not both score high with the corpus present. Second, and this is what makes the contrast trustworthy, we *calibrate* necessity against a constructed weight-level baseline (Section 4) that these methods have no analogue for.

The closest relative, and the structural gap. **ContextCite** [Cohen-Wang et al., 2024] is the nearest method: because it ablates context and measures the change in the response's log-probability against the model's own baseline, it *partially* tracks necessity. But it requires open weights and per-token log-probabilities (so it cannot run on the closed API models our test does), attributes text-influence rather than a task outcome, carries no weight-level known-groups calibration, and yields no relied/redundant/unusable/misled split. The gap common to this whole strand is that none has a *constructed parametric baseline*: a way to teach a fact into the weights and confirm the measure responds. That construction (Section 4) is what lets us tell a fact the model relied on from one it merely already knew—and, in the misled case, a fact it faithfully followed off a cliff.

Knowing vs. doing. A separate line finds that knowledge a model holds need not govern its behavior: facts learned in one form fail to surface in another [Berglund et al., 2023], edits leave messy or mis-scoped ripples [Cohen et al., 2024], stated reasoning need not reflect the true cause of an answer [Turpin et al., 2023], and tool-use exhibits a measured knowing–doing gap [Anonymous, 2026]. We connect this strand to *necessity measurement*: because our instrument scores a decision,

Policy	mean J	median	min	max	cleared
naive (hold course)	1765.44	1988.08	1.08	1995.18	11%
VO reference	0.70	0.58	0.33	1.27	100%
SB-MPC reference	0.01	0.01	0.00	0.03	100%

Table 1: Construct validity and grading across nine varied encounters. The objective J separates the three competence levels cleanly and takes a continuous range of values (20 distinct across all cells, $[0.00, 1995.18]$)—a graded transfer metric, not a near-binary check.

the same gap becomes a validity condition on the measure—a taught fact that is recallable can leave the scored decision unmoved, so necessity read off a single-shot decision under-reports it (Section 4). Unlike Anonymous [2026], which reads internal cognition against tool-call execution, our axis is a counterfactual ablation of a *corpus item* against a simulator objective, and our construction lets us show the gap closes under elicitation but not under direct fine-tuning.

3 The transfer instrument

The necessity measure is a with-versus-without contrast on a *task outcome*; this section defines that outcome and shows it grades and isolates cleanly, which is what licenses the contrast in Section 4.

Task and metric. The worked domain is COLREG collision avoidance: a power-driven vessel must choose course and speed to resolve an encounter. We implement a kinematic simulator with an elliptical ship domain and a Collision Risk Index, and two reference solvers—velocity obstacles [Fiorini and Shiller, 1998] and simulation-based MPC [Johansen et al., 2016]. For a learner policy π on scenario set S we define the transfer outcome as regret against a reference policy $\pi^\star$,

$$\mathcal{R}(\pi) \;=\; \frac{1}{|S|}\sum_{s\in S}\big[\,J(\pi,s)-J(\pi^\star,s)\,\big], \tag{1}$$

where J is the per-encounter objective (a weighted sum of a safety-margin barrier, COLREG compliance penalty, and route deviation; lower is better). We deliberately say "reference," not "optimal": VO and SB-MPC are strong near-optimal solvers but not proven globally optimal for the full non-convex encounter. The instrument needs only a fixed, reproducible reference that construct validity shows separates naive from expert behavior; we report which solver is used.

Construct validity and grading. The instrument must separate good from bad policies before any necessity claim, and—to be a transfer measure rather than a pass/fail check—the metric must *grade*, not collapse to two values. Across nine varied encounters (head-on, starboard/port crossing, overtaking, restricted-visibility geometries), three policies of increasing competence separate cleanly and continuously on J (Table 1): a naive hold-course policy (mean $J \approx 1765$, clearing 11% of domains), a velocity-obstacle reference ($J \approx 0.70$, all clear), and SB-MPC ($J \approx 0.01$). The objective takes 20 distinct values in $[0.00, 1995.18]$ over the policy$\times$scenario cells—a graded metric, with the safe regime itself spanning a continuous 0.00–1.27 range. This is the novice/expert-separation logic of Van Dongen et al. [2007], applied to a decision/control task.

KC$\rightarrow$single-metric identifiability. For the diagnosis to attribute a task failure to a *specific* corpus rule, each knowledge component must govern exactly one metric, so ablating a KC degrades that metric and no other. This is a design requirement to be verified per instrument, not a tautology: designing the metrics does not guarantee that ablating one KC leaves the *others* unchanged—if the objective's terms are entangled, ablation bleeds across metrics. Identifiability is the empirically-checked absence of that bleed. On a discrete procedural domain (a roadside tire change) the verification is clean (Table 2): each ablation degrades only its governed metric, maximum off-diagonal change 0.0000. Because the property holds on both a continuous-control and a discrete-procedural instrument, it is a general measurement-design property, not a COLREG artifact. Estimator-recovery and calibration on synthetic data appear in Appendix C.

ablated KC ↓ / metric →	safety	core	sequencing	finishing
safety	$-\mathbf{1.00}$	0.00	0.00	0.00
core steps	0.00	$-\mathbf{1.00}$	0.00	0.00
sequencing	0.00	0.00	$-\mathbf{0.22}$	0.00
finishing	0.00	0.00	0.00	$-\mathbf{1.00}$

Table 2: KC→single-metric identifiability (procedural domain). Change in each metric after ablating each knowledge component. The diagonal (bold) is the intended governed metric; every off-diagonal cell is 0.00 (max $|\Delta| = 0.0000$)—ablation does not bleed across metrics.

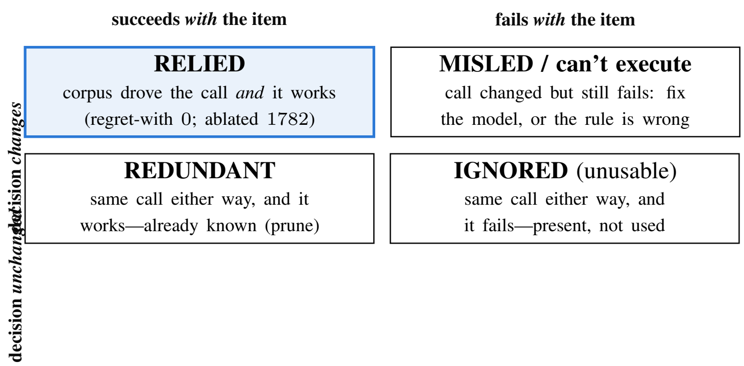

Figure 1: Necessity read as *different*×*usable*: ablate an item and ask whether the *decision* changes (rows) and whether the learner *succeeds with* it (columns). Only the top-left cell is **relied on**—the corpus changed the call, for the better. The top-right cell is the *misled*/can't-execute case a subtractive "did removing it hurt" test cannot see. A coarser categorical verdict is deferred to Appendix A.

4 Necessity, calibrated by construction

This is the headline. We first fix how necessity should be *read* off the instrument, then make the reading trustworthy by constructing its ground truth.

Necessity as *different and better*. Ablate a corpus item and compare the run against the with-corpus run on two observables (Figure 1): does the *decision* change (**different**), and does the learner *succeed with* the item present (**usable**)? An item is **relied on** only when the corpus changes the decision *for the better*—removing it flips a succeeding call into a failing one. Reading necessity as "the corpus changed the call, and it helped" is stronger than the subtractive "nothing broke when we removed it," and, crucially, the two-observable reading is what will separate genuine reliance from the *misled* case in Section 5: a corpus item can change the decision hard (high raw necessity) while the learner still *fails* with it present, which the subtractive test cannot see. This reading is computed off the graded regret instrument itself, not a decoupled side-metric.

Why the reading needs calibration. The necessity test infers a fact about a model's *weights* from a change in the *prompt*: ablate an item, and if the governed outcome does not move, the model did not need it. The sharpest objection is that context removal is not weight-level knowledge change, so a small necessity could be a no-context heuristic in disguise, or a large necessity could reflect a fact the model happens not to know for reasons we did not control. We close this *by construction*: if teaching a fact into the weights drives necessity for that fact down as a dose-response, then a near-zero reading is evidence the model already knew the item, and the measure is a non-confounded, known-groups quantity. We build models along a gradient from fact-naive to fact-knowing by teaching a target fact into the weights (LoRA SFT) and sweeping two independent knowledge gradients—the LoRA scale α (one adapter, evaluated at $\alpha \in [0, 1]$) and training checkpoints—so their agreement rules out a gradient-method artifact. We establish the calibration in a simulator-free QA domain, then use a decision-based domain to surface a limit the calibration alone would hide.

	α=0	0.25	0.5	0.75	α=1
Fictional-fact QA necessity, Qwen2.5-3B	1.00	1.00	0.48	0.04	0.05
Fictional-fact QA necessity, Phi-3.5-mini	1.00	1.00	0.93	0.13	0.01

Table 3: Necessity dose-response by construction (the calibration), mean over three seeds. Teaching many independent invented facts into the weights drives the instrument's necessity for them smoothly down the LoRA-α gradient in *two* model families—monotone in every seed, differing only in where the knee falls (the Qwen α=0.5 point spans 0.39–0.56 across seeds). That two architectures agree rules out a single-model artifact. The answer-accuracy objective uses no simulator. The decision-based hazard domain does *not* yield such a curve—teaching the fact as prose leaves the scored maneuver unmoved (the *knowing* $\neq$ *doing* gap of Section 4); it is reported there, not as a second calibration.

The calibration: simulator-free fictional-fact QA. We invent facts about fictional entities (research outposts and their attributes) that no model can have memorized, and score answers with a programmatic checker—no simulator, no reference policy. Because the facts are fictional, a competent model is necessarily corpus-bound at baseline, and because there are many *independent* facts, partial teaching learns a fraction of them, so the aggregate necessity is *graded*. Teaching the facts into *two* independent model families drives mean necessity smoothly and monotonically down the LoRA-α gradient: $1.00 \rightarrow 1.00 \rightarrow 0.48 \rightarrow 0.04 \rightarrow 0.05$ for Qwen2.5-3B and $1.00 \rightarrow 1.00 \rightarrow 0.93 \rightarrow 0.13 \rightarrow 0.01$ for Phi-3.5-mini (three seeds each, monotone in every seed; Table 3). That two architectures give the same collapse—differing only in where the knee sits—rules out a single-model artifact, and the graded floor near zero is what a near-zero reading on the real instrument is calibrated against. This is the known-groups calibration, in a domain we did not build a simulator for—answering the objection that the measure is an artifact of one hand-built environment. (Necessity here is measured closed-book: the without-corpus condition permits parametric answering, so taught knowledge surfaces rather than being suppressed by a strict "answer only from the provided facts" instruction.)

A decision-domain caution: knowing $\neq$ doing. The calibration teaches and scores in the *same* channel—a fact, queried as a fact. A behavioral instrument scores a *decision*, and there the two can come apart, so we probe the decision domain directly with a corpus-only charted hazard (a danger scored by the objective barrier but not shown in the prompt, knowable only from the corpus) taught into a Qwen2.5-3B model *as prose*. The fact becomes *recallable*—free-text recall of the avoiding action rises from 0.00 to 0.78—yet the scored maneuver *does not move*: $\mathcal{R}$-necessity is flat (≈ 667 at both the naive and taught ends), and stays flat across both the LoRA-scale gradient and a genuine partial-training checkpoint sweep (three seeds; one seed shows a transient turn only at deep intermediate checkpoints, and the fully-taught adapter returns to 667—an instability, not a reached decision). The fact is in the weights but does not reach the decision: a *knowing* $\neq$ *doing* gap, a behavioral cousin of the reversal curse [Berglund et al., 2023] and of the knowing–doing gap reported for tool use [Anonymous, 2026]. Two interventions localize it (Figure 2). *Eliciting* the knowledge by reason-then-decide prompting closes the gap—necessity falls to ≈ 0.3, while a naive-model reasoning control still grounds, so it is the taught knowledge being surfaced, not reasoning alone. *Installing* the decision directly by fine-tuning on the scored format does not: across positive-only, contrastive, and surface-debiased teach sets the model learns a blanket or shortcut policy—confabulating the hazard at every location, an instance of unfaithful reasoning [Turpin et al., 2023] and of mis-scoped knowledge editing [Cohen et al., 2024]—never the conditional fact (Appendix B reports the parallel weight-level *says* $\neq$ *does* dissociation from unlearning). The lesson for the measure: on a decision objective, whether a taught fact "counts" depends on whether it is *elicited*, so necessity read off a single-shot decision can under-report knowledge the model in fact holds—a validity caveat we return to in Section 6.

5 The corpus diagnosis

The calibrated necessity measure is the input to a per-item *corpus diagnosis*: run the different$\times$usable reading (Figure 1) on every item and label it. The three non-relied cells demand different fixes. An *unchanged* decision that succeeds is **redundant** (the model already knows it;

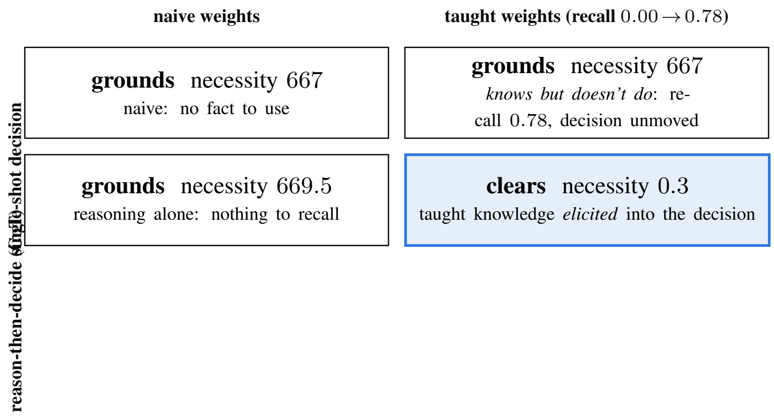

Figure 2: The knowing$\neq$doing gap and its bridge, on the hazard decision ($\mathcal{R}$-necessity; lower = clears without the corpus). The taught fact is recallable (top-right, recall $0.00 \to 0.78$) yet leaves the single-shot decision unmoved (top row, flat 667). Only *reason-then-decide on the taught model* clears (bottom-right, 0.3); reasoning on the naive model does not (bottom-left), so the turn is the taught knowledge being elicited, not reasoning alone—the action needs *both* the knowledge and the elicitation. Installing the decision directly by fine-tuning does not reproduce this: it yields a blanket/confabulated policy (`decision-teach-arc.md`), not the conditional fact.

prune it), calibrated against the dose-response floor (Section 4) so a near-zero reading means already-known rather than never-used. An unchanged decision that fails is **ignored**/*unusable* (present but the model cannot execute it). The remaining cell is the sharp one.

The misled result. The top-right cell of Figure 1—the decision *changes* but still *fails*—is where a learner faithfully follows a *wrong* retrieved rule. This is the failure the whole measure is built to expose. Consider a corpus item that instructs the wrong give-way maneuver. A faithful learner applies it, and when the item is ablated the learner falls back on a safer parametric reflex and clears—so the item's raw necessity is *maximal*: the decision swings hard, exactly the signature of strong reliance. Yet the outcome *with* the item is a grounding. A subtractive necessity test ("did removing it hurt?") and an influence-based faithfulness score both report this item as heavily used and well-grounded—they cannot distinguish a learner relying on a correct rule from one faithfully executing a harmful one, because both change the output and both are "supported" by the context. The two-observable reading separates them: the misled item lands in the top-right cell (changes the call, fails with it), never in **relied** (top-left). The instrument's diagnostic value is precisely that faithful compliance with a wrong rule is worse than ignoring it, and only the succeeds-with observable can see the difference.

The learners that exercise the override. Necessity is always measured against a *learner*, and the misled/override contrast needs a cast that varies in how deeply it uses the corpus: a **blind** learner cannot perceive the danger and collides; a **naive** learner is sighted but untrained and avoids generically; an **implementer** applies a corpus rule verbatim where the situation matches; and a **reasoner** treats the corpus as premises and combines it with base rules for novel cases. On a matched suite where the textbook rule already fits, implementer and reasoner are indistinguishable (Δregret 0.00); on a conflict suite built so the safe side *alternates* port/starboard—so no fixed reflex passes—the reasoner clears $6/6$ while the lookup implementer clears only the 3 reaches where the local rule is redundant with Rule 14 and *grounds on the* 3 *it overrides*. The necessity of the local-rule corpus is thus $3/6$, separable *only* on the overriding cases: this is the RAG-as-lookup vs. RAG-as-reasoning-substrate distinction made measurable, and it is exactly the axis on which a *wrong* local rule produces the misled failure above (an implementer follows it off the cliff; the two-observable reading catches the result). A live model run confirms the reading is not an artifact of hand-built policies: Gemini-2.5-flash applies the local rule like a reasoner, relying on it on all three override reaches ($5/5$ samples, necessity 1782 ± 0)—with the corpus it alters to port and clears, ablated it applies Rule 14 and grounds.

Model	relied	unusable	redundant
Claude Opus 4.5	**8**	0	0
Claude Haiku 4.5	4	0	4
Claude Sonnet 4.5	3	$2^{\dagger}$	3
Amazon Nova-micro	$2^{\dagger}$	$2^{\dagger}$	4
Llama-4-Maverick-17B	0	4	4

Table 4: Cross-model necessity over the eight-hazard corpus-only suite (five models, AWS Bedrock). Each hazard is isolated in its own corpus, so the three counts partition the suite of 8 and per-model reliance is a fraction. Classification uses *necessity* $= \mathcal{R}(\text{ablated}) - \mathcal{R}(\text{with-corpus})$ and *regret-with*: **relied** $=$ necessity > 100 *and* regret-with < 10; **unusable** $=$ regret-with ≥ 10 (grounds *with* the item present); **redundant** $=$ otherwise (clears both ways). Values are the temp-0 point estimate; the $K{=}5$ temp-0.7 ensemble has sd ≈ 0 except (†): Sonnet unusable 1.2 ± 0.7, Nova-micro relied 0.2 ± 0.4 and unusable 2.2 ± 0.7. **Counts reflect ablated-arm fallback as well as reading:** reflex-appliers cap at $4/8$ (the four override reaches), and Opus's $8/8$ reflects that it holds course rather than falling back on Rule-14 when the rule is ablated.

The misled result on a real model. The same live model, run through a *misled* arm—a corpus rule that instructs the wrong (port) give-way toward an *observable* obstacle, so following it grounds while the ablated model clears—makes the blind spot empirical rather than hypothetical. Over four misled scenarios $\times$ six samples (temp-0 point $+$ $K{=}5$ at 0.7; $n{=}24$ draws per condition), Gemini's harmful into-the-obstacle maneuver rate rises from $\mathbf{4}\%$ ablated to $\mathbf{33}\%$ with the wrong rule, while its correct give-way rate falls from 70% to 33%—the corpus roughly $8\times$'s the harmful action and halves the correct one. Read per sample as a binary cell the signal is noisy (misled on $1.6/4$ cells on average); read as the *marginal maneuver rate* it is a clean, large shift, and the marginal rate is what we report. The mechanism is not a perception failure—ablated, the model almost never turns toward the obstacle ($1/24$ draws) and correctly cites it—but a *confabulation*: with the wrong rule present it turns toward the obstacle and reports that doing so "passes clear of the moored barge." So the model that applies *correct* local rules like a reasoner (above) is, on *wrong* ones, a rule-follower whose harmful compliance a standard necessity or faithfulness score would credit as high, well-grounded reliance—exactly the case the succeeds-with observable is built to catch. We report this as a single-model effect size and its mechanism, not a rate we generalize.

Offline recovery of known ground truth. Before any real-model sweep we confirm the classification recovers truth we control. Against two reference mock learners the instrument is exact: the corpus-bound mock's decision *changes* on ablation and it then grounds (regret 0.2 with the rule $\rightarrow$ 1981 without $\Rightarrow$ *relied*); the leaking mock's decision is *unchanged* and clears both ways ($\Rightarrow$ *redundant*). The whole offline backbone regenerates in one deterministic, LLM-free command that checks each headline result against its claimed value.

A supporting cross-model reading (demoted). As one supporting breadth result we run the necessity test as a suite of eight independent, alternating-bow charted hazards, each isolated in its own corpus, so per-model necessity is the fraction of the suite relied upon (Table 4). This gives a graded reliance spectrum across five production models, but the counts must be read with an important caveat: they reflect ablated-arm fallback as well as reading. The suite is 4 override reaches (the hazard sits where the default Rule-14 starboard turn grounds) plus 4 where starboard already clears, so a model that falls back on the starboard reflex when the rule is ablated is *rescued* on the four non-override reaches and caps at $4/8$ *by construction*. Claude Opus's $8/8$ is co-determined by this: it *holds course* when ablated (strict-mode faithfulness—"no rules were provided") rather than falling back on Rule-14, so it grounds on all eight, whereas reflex-appliers like Haiku (4) cap at four by construction. Opus's $8/8$ therefore means it does not fall back, not that it out-reads Haiku; the spectrum grades reading and ablated-arm conservatism together, which is why we demote it to support rather than headline.

Per-rule attribution. To show the diagnosis attributes reliance to a *specific* rule rather than to corpus presence in general, we add a second, independent rule (Rule 15 crossing give-way) exercised on its own crossing scenarios. Ablating each rule produces a large regret signal on its own encounters and localizes there: on `gemini-flash-latest`, ablating Rule 14 gives $\delta_{\mathcal{R}}{=}1981$ (head-on) and ablating Rule 15 gives 1285 (crossing), each on its own disjoint scenario subset, so the instrument reads reliance per rule, not merely per corpus.

Corpus sufficiency and *false sufficiency*. Rolling the per-item classifications to the corpus level yields a sufficiency judgement for a query set: *contributing* (every item relied upon), *partial*, *unusable*, or—the diagnostic ordinary RAG metrics cannot produce—*false sufficiency*: task accuracy is high yet necessity is ≈ 0 everywhere, so the corpus is not carrying the load and success rides on priors, fragile to distribution shift. The claim is explicitly scoped to the probed query set (sufficiency is always "for these queries"); the false-sufficiency signal is precisely the measurable warning that a bounded-query sufficiency claim will not survive a shift. A finer presence-vs-retrieval decomposition is developed in Appendix D.

6 Limitations and scope

Mechanism, not external validity. Every result here is simulation-based mechanism evidence. We make no learning-gains claim; the pre-registered human cohort study is the external-validity test this in-silico screen is built to de-risk, and it is the endpoint, not a claim of this paper. **Single models and seeds.** The calibration dose-response and the knowing $\neq$ doing decision-domain results are single-model, single-seed runs; a queued multi-seed GPU rerun (≥ 3 seeds, and a second model family) is documented in `experiments/provenance/gpu-pending.md` and `decision-teach-arc.md`, and until it lands we treat the curves as single-run and keep the hedging in Section 4. The offline mock recovery already confirms the *instrument* recovers ground truth on these domains; what is pending is the multi-seed *taught-model* band. **Small evidence base.** The cross-model reading (Table 4) covers five models across three provider families on an eight-hazard suite, and reflects ablated-arm fallback as much as reading (Section 5); it is a worked example, not a leaderboard. **Decomposability may not generalize.** KC$\rightarrow$single-metric identifiability and clean per-rule ablation are properties we *design into* a governed corpus and *verify* per instrument (Table 2); an arbitrary corpus whose rules are entangled may only support coarser, group-level attribution. **Which channel counts is a construct choice.** Necessity is read through a *channel*—a recalled fact or an executed decision—and the knowing $\neq$ doing gap (Section 4) shows the two can diverge: a fact the model demonstrably holds (recall rises $0.00 \rightarrow 0.78$) can leave a single-shot decision unmoved. The valid channel is therefore a property of the downstream question, not of the instrument. For a corpus-bounded *tutor*, whose target is what the learner comes to *know*, the recall/knowledge channel (our fictional-fact calibration) is the faithful proxy, and reading necessity off an un-elicited decision would wrongly score an acquired fact as *unnecessary*; for a corpus-bounded *agent* that must act, only the decision counts and a knowing-but-not-doing fact is a genuine failure the recall channel would miss. An instrument should declare its target construct and, where application matters (procedural knowledge, Koedinger et al., 2012), elicit before scoring or measure both channels and treat their gap as diagnostic rather than noise. **Proxy validity.** Simulated learners can be unfaithful; recent work documents systematic LLM-vs-human fidelity gaps [Yuan et al., 2026, Wu et al., 2025, Srivatsa et al., 2025], which is another reason the human study, not the screen, is the endpoint.

7 Conclusion

We described an in-silico instrument that measures whether a learner *needs* a corpus before any human-subject effort, and made the measure trustworthy by *constructing* its ground truth: teaching many invented facts into a model's weights drives the instrument's necessity for them down as a graded dose-response ($1.00 \rightarrow 0.03$, in a simulator-free QA domain, with LoRA-scale and checkpoint gradients agreeing). A decision-based domain adds a validity caution the field should carry: a fact taught as prose is recallable yet does not move a single-shot decision (a *knowing* $\neq$ *doing* gap) unless reasoning *elicits* it, so the necessity channel must match the downstream construct. On that calibrated measure we built a corpus diagnosis whose sharpest output is the *misled* case—a learner faithfully following a wrong retrieved rule looks maximally reliant yet is harmed, invisible to influence-based faithfulness and to subtractive necessity, and caught only by reading necessity as decision-change *and* succeeds-with. The claim is testable and narrow, the domains are worked examples, and the measurement method—calibrated, and honest about being a pre-human-study screen—is the point.

A The categorical leakage verdict (secondary)

The main text classifies each item on two observables—does the decision change on ablation, and does the learner succeed with the item present (Figure 1). For continuity with prior leakage work we also compute a coarser *categorical* verdict, CORPUS-BOUND/LEAKING/INCONCLUSIVE, by a majority vote over three signals: the ablation-delta (does regret rise when the rule is removed), a *counterfactual* test (replace "alter to starboard" with "alter to port"—does the learner follow the altered rule?), and a *closed-book* contamination baseline (no corpus supplied). The majority vote keeps a single dissenting sub-test from vetoing a clear ablation-and-contamination pattern. We de-emphasize it because it is noisier than the two continuous observables and *undercounts* a model whose decision plainly depends on the fact but which answers closed-book from priors—it put Haiku at $0/8$ where the decision-change reading gives $4/8$.

Strict-mode reading on standard COLREG. The verdict is retained mainly for one signal it reads cleanly. On *standard* COLREG, Claude Opus verdicts CORPUS-BOUND, but the audit log shows this is strict-mode faithfulness, not a knowledge gap: when the steering rule is ablated Opus abstains/holds course ("no rules were provided…") rather than falling back on its obvious parametric Rule-14, while Haiku falls back to Rule-14 starboard (LEAKING) and Sonnet splits across samples (INCONCLUSIVE). This is the same ablated-arm behavior that co-determines Opus's $8/8$ in Table 4.

B Validation by unlearning: a *says* $\neq$ *does* dissociation

The complementary weight-level direction removes a rule the model *does* know and asks whether necessity rises. This arm is secondary—it has dynamic range only on a rule with a real corpus-vs-prior gap, which the standard rules lack—but it yields an independently interesting result. We remove the alter-to-starboard knowledge from an open-weight model with a real unlearning method—SimNPO [Fan et al., 2024] as primary (reference-free, length-normalized), with NPO [Zhang et al., 2024], RMU [Li et al., 2024], and gradient ascent [Jang et al., 2022] as baselines—and score the base vs. unlearned model on the reference-policy instrument. A removal audit [Lynch et al., 2024] reports whether the knowledge is *gone* vs. merely *suppressed*; for a stable novice that resists benign relearning we layer in a robust utility-preserving recipe [Singh et al., 2025]. A completed GPU run (gentle, guardrailed SimNPO on Qwen2.5-3B; config in Table 5) delivers a lesson sharper than the predicted 2×2: **standard free-text unlearning metrics overstate removal because they measure what the model *says*; the reference-policy instrument measures what the model *does*—its steering decision—and shows the decision is untouched.**

Primary result: a *says* $\neq$ *does* dissociation. Every metric that reads the model's *words* registers forgetting: on the GPU run the forget-probe "starboard" rate falls $0.43 \to 0.27$, the forget-set NLL rises $7.4 \to 32.4$ (retain-set NLL, teacher-forced, even *improves* $4.3 \to 1.07$), and—most tellingly—the Rule-14 *citation* disappears, the model reattributing its maneuver to Rules 15/16/17. (A CPU run reproduces the likelihood collapse method-agnostically: forget NLL $4.6 \to 92$, direction-cue $5/6 \to 0/6$.) Yet the *decision* the instrument scores does not budge: the unlearned model's completions *still turn* $+30°$ *to starboard* (model damage flat at 0.10, retain-coherence 0.91). A words-level metric reports a fact removed while the policy that governs behavior is entirely intact.

Suppressed, not removed; and the aggressive inversion. Twenty benign fine-tuning steps pull the forget-set NLL $32.4 \to 9.2$, back near its base value—the textbook suppressed-not-removed signature. By contrast an earlier aggressive recipe (lr 1×10^{-4}, no guardrail) drove model damage $0.10 \to 0.33$ and turned the ship to *port*—the wrong way—on $6/8$ prompts: because the head-on turn is essentially binary, the default objective simply relocated the mass onto "port," *inverting* the safety rule, a caution that naive weight-level unlearning of a safety rule can yield confident non-compliance. **Why the 2×2 is flat:** Rule 14 already LEAKS at baseline—every frontier model knows head-on $\to$ starboard parametrically—so ablating it changes nothing (ablation-delta 0), leaving no dynamic range. The LEAKING verdicts are therefore *correct*; this is exactly why the primary weight-level validation is the construction dose-response (Section 4), where a corpus-only fact *does* have range.

Metric	1.5B (CPU, SimNPO) base	1.5B (CPU, SimNPO) unlearned	3B (GPU, gentle SimNPO) base	3B (GPU, gentle SimNPO) unlearned
Forget-target NLL (↑)	4.6	92.0	7.4	32.4
Retain-set NLL, teacher-forced (↓)	3.2	0.18	4.3	1.07
Forget-probe "starboard" rate (↓)	5/6	0/6	0.43	0.27
Retain-coherence, free-gen (↑)	—	—	1.00	0.91
Model damage (wrong+degen., ↓)	—	—	0.10	0.10
Steering decision (instrument)	—	—	*stbd*	*stbd*

Table 5: Gentle, guardrailed SimNPO unlearning of the alter-to-starboard rule, on a 1.5B model (CPU) and Qwen2.5-3B (GPU, `bfloat16`; lr 5×10^{-5}, retain_weight 3.0, Rule-14 kept in the retain set). Every *words-level* metric registers forgetting while retain utility is preserved and there is no inversion (model damage flat at 0.10). Yet the *decision* the instrument scores is untouched: the unlearned model still turns $+30°$ to starboard, and the 2×2 stays verdict LEAKING, ablation-delta 0.000—a says $\neq$ does dissociation. Benign relearning (20 steps) pulls forget NLL $32.4 \to 9.2$, confirming the fact was *suppressed*, not removed.

C Estimator recovery and calibration

Feeding the instrument's outcomes into a generic learner-agent estimator, we recover known ground truth on synthetic data (Table 6). This turns "we have a metric" into "the measurement is valid."

Estimator	Recovery of known ground truth	Calibration (ECE)
Elo	static ability, mean abs. error 0.046	0.023
BKT	latent state $P(\hat{L}) = 0.997$ (learned) vs. 0.175 (not)	0.004

Table 6: Estimator-recovery and calibration on synthetic data with a known ground truth. The measurement, not just the metric, is validated.

D Presence vs. retrieval: what in the corpus changes the answer

The necessity test asks whether the learner relies on the corpus *at all*; a deployed RAG tutor raises a finer question—of the value curation adds, how much comes from the fact being *present* versus the retriever actually *surfacing* it. Decomposing over in-corpus $\times$ retrieved gives three scorable conditions: NONE (closed-book, priors only), RETRIEVED (the deployed dense retriever's top-k), and ORACLE (the answer-bearing chunk, i.e. perfect retrieval). Accuracy across them attributes the gain: $\mathrm{acc}(\text{ORACLE}) - \mathrm{acc}(\text{NONE})$ is the *content ceiling*, $\mathrm{acc}(\text{RETRIEVED}) - \mathrm{acc}(\text{NONE})$ is what the deployed pipeline *delivers*, and their difference is the *retrieval gap*. On seven equipment-SOP fact-recall queries through a small local generator (Qwen2.5-1.5B), closed-book accuracy is $0/7$, ORACLE is $7/7$, and the deployed MiniLM pipeline lands between at $3/7$—so of a 100% content ceiling the pipeline *delivers* 43% and *forfeits* a 57% retrieval gap. A curation failure (fact absent), a retrieval failure (present but unranked), and a priors success are distinct outcomes the binary leakage verdict cannot separate. (Numbers illustrate the method—one small model, seven queries—not a benchmark.)

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